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Technology Trends in Indian Stock Exchanges: A Study of Mode of Trading at NSE

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ABSTRACT

Stock Market Trading has grown from manual to electronic to automated with the use of sophisticated technology in the last decade. The increased adoption of technology will lead to disruptive trends in the current decade in stockexchanges arena.The tremendous growth in financial trading volumes in the last couple of decades has been made possible only with the use of technology.This paper attempts to analyze the key trends in trading technologies in Indian Capital Markets of 21st Century. In particular the mode of trading at NSE is examined. In this connection Algorithmic, Non-Algorithmic, Direct Market Access (DMA), Co-Location, Internet Based Trading (IBT), Mobile Trading, Smart-order Routing (SOR) are analyzed extensively.

KEY WORDS: *NSE, Algo, Non-Algo, DMA, Co-Location, IBT, Mobile Trading, SORJEL Classification:G1, G11, G12, G14*

1. INTRODUCTION:

Technology has been the prime driver of finance sector around the world and is a key enabler of economic development in India. Technology has also played a great role in the evolution of stock exchanges. Today people can access, trade and invest much more easily than before. The securities market today enjoys an extended reach and serves investors in remote areas also. Over the past few decades, technology has transformed the trading of securities. Before the arrival of computers into Stock Market, all trading was conducted in pits or trading floors.

Back offices were filled with clerks to ensure that transactions were properly completed and reconciled. Gradually both the back office and the actual trading process have been transformed by automation. Many financial markets have abandoned floor-based trading systems in favor of electronic limit order book or Screen based trading systems.There was much debate about moving from floor-based trading systems to screen based trading systems (SBTS) in the 1970s. SBTS has become accepted as the pre-dominant form of trading only in the last two decades.

In India, there has been a major shift in the way business is done in the Stock market. There has been a long journey from meeting under the banyan tree in Mumbai to hiding behind computer screens and executing trades at the speed of light. Indian Securities Markets witnessed huge transformation through technology since 1990s. Rapid advances in IT have determined important changes and innovation in the operation of stock exchanges. Technological advancements in the Indian Securities Market transformed the way trading takes place and has in turn, resulted in the process becoming easier, faster, transparent and safer for the investors.

Historical Background:

Pre-1994	Post-1994
1. Low Transparency	Full Transparency
2 Low Reach	Nationwide Reach
3 Weak Risk Management	Real –time Risk Management
4 Long Clearing & Settlement cycles	T+2 Clearing & Settlement cycles
5 Paper Intensive	Electronic Matching
6 Physical Share Certificates	Full De-Materialization
	F&O, Currency, Interest Rate Futures
	Real-Time Cross Margining

Source: NSE

2. LITERATURE REVIEW:

There are extensive studies, reports and books available on Indian Stock market and Technology.We have reviewed some literature pertaining to the topic under study. The relevant literature is reviewed on basis of books, periodicals, newspapers and websites. The detailed review is given below.

Cardella, et.al (2014)in their study entitled "Computerization of the Equities, Foreign Exchange, Derivatives and Fixed Income Markets" surveyed empirical studies on the development and effects of increased computerization across various financial markets. The researchers find that there have been dramatic changes in certain derivative markets, foreign exchange and in particular equity markets. The research has found positive effects of computerization on measures of market quality, but the survey highlights that human intermediation is still prominent and beneficial in certain areas. In a report entitled " The future of computer trading in financial markets" published by Government office for Science reports that there is no direct evidence that high frequency computer based trading gas increased volatility.

Goldstein, Kumar & Graves (2014)in their article titled "Computerized and High-Frequency Trading" report that the use of computers to execute trades has increased overtime,resulting in a variety of computer algorithms executing electronically targeted trading strategies at high speed. The researchers describe the evolution of increasingly fast automated trading over the past decade. The authors survey and contrast several studies on the impact of high-speed trading on the performance of securities market. The authors examine some of the regulatory questions surrounding the need for safe-guards over the fairness and risks of high-speed, computerized trading.

Lin (2013)in his article entitled "The New Investor" discusses the sea changes happening in finance. The researcher emphasizes that machines are on the rise and humans are on the decline. Technological advances have made finance faster, larger, more global, more inter connected and less human. According to the researcher modern finance is becoming an industry in which the main players are no longer entirely human. Instead the key players are now cyborgs: part machine, part human. Modern finance is transforming into what this article calls cyborg finance. This report further opines that over the last quarter century, computerization and artificial intelligence have revolutionized finance and they continue to fundamentally transform finance from an industry dominated by humans to one in which humans and machines share domination. The authors further opine that this transformation resulted from advances in technology and regulatory reforms over the last few decades.

Bhunia & Ghosal (2011) in the paper titled "An impact of ICT on the growth of capital market –empirical evidence from Indian Stock Exchange" investigate the impact of ICT on the growth of the Indian Stock Exchange using a modified version of the Gompertz technology diffusion

model introduced by Chow (1983) and accordingly reshuffles the model with ICT development. The results of the study reveal that preferred variables are appreciably affected by information and communication technology more than ever in respect of amplifying the number of stock brokers, investors and admittance to ICT.

Ming-Chi Lee (2009) in a study entitled "Predicting and explaining the adoption of online trading: An empirical study in Taiwan" investigates howstock investors perceive and adopt online trading inTaiwan.We developed a researchmodel which integrates perceived risk, perceived benefit and trust, together with technology acceptance model(TAM) and theory of planned behaviour (TPB) perspectives to predict and explain investors' intention to use onlinetrading. The model is examined through an empirical study involving 338 subjects using structural equationmodelling techniques.

Gorham & Singh (2009)in their book titled "Electronic Exchanges-The Global transformation from pits to bits" present a history of exchanges,their purpose and development. They delve into the process of floor-based trading and the technological innovations that led to the development of electronic exchanges.The authors identify four major themes in the transformation of exchanges: 1) the shift from floors to computer screens 2) the shift from private clubs to public companies 3) the shift from local and national to Global competition 4)the shift from smaller to larger operations. According to the authors, the most important of all the transformations that exchanges are undergoing is the shift from trading floors to trading screens.

Report on Indian Exchanges (2009) by IDFC research describes that transparency, annuity revenues; high operating leverage and solid entry barriers make exchanges a near-perfect business. The report further argues that Indian Exchanges are almost on par with global peers in terms of corporate structure and sophistication of systems, Indian Exchanges are gathering scale. As elucidated in the report exchanges have been around for centuries, but the business model continues to strengthen with the industry increasingly gaining depth.

Shah, et.al (2008) in the book titled "India's Financial Markets" observes that the public equity market within India, both spot and derivatives, takes place almost entirely at the two exchanges – NSE and BSE. There is an open electronic limit order book (ELOB) with order matching by the trading computer. The author reports that the processes of organized financial trading in India have focused on Exchanges.John (2007) describes the rapid technological change that characterized Indian financial markets in the last three decades of twentieth century by increasing the opportunity costs of maintaining India's unreformed equity market microstructure. The author argues that India eventually adopted many latest innovations, leapfrogging from archaic market institutions and practices in the early 1990s to international best practices at the beginning of new millennium. The paper further elucidates that by 2001, reforms brought India up to par with the global standards for every aspect of its

equity market microstructure.Jain & Johnson (2006)in their article titled "Trading Technology and Stock Market Liquidity: A Global Perspective" expound on technological revolution that swept financial markets around the world. Computerization and satellite communication have transformed the organization of stock exchanges and improved secondary market liquidity. Lee (2002)in the paper titled "The future of securities exchanges" presents a range of predictions about securities exchanges. The author analyzes that historically, exchanges have had seven main types of revenues: 1) Membership subscription, fees for 2) Listing, 3) Trading 4) Clearing and 5) Settlement and charges for the provision of 6) Company news and for 7) quote and trade data. In the paper four broad themes related to securities exchanges are discussed concerning information, industry, governance and politics.

3. OBJECTIVES OF THE STUDY

3.1) To characterize in detail the nature and architecture of trading technology in Indian Stock market

3.2) To analyse mode of trading in National Stock Exchange of India

4. TECHNOLOGY FOR ACCESSING EXCHANGES IN INDIA

The current landscape of trading access in Indian Markets can be summarized as below:

a) Manual Order entry through exchange terminals (NEAT,BOLT)

b) Order Management Systems (CTCL)

c) Direct Market Access for Sophisticated Clients (DMA)

d) Internet Trading for Retail Investors (Via PCs, Mobile and Wireless platforms)

e) Co-Location Facilities (Co-Lo)

- For proprietary traders and
- Brokers offering lower latency access to clients

f) Algorithms for proprietary trading

- High Frequency Trading Algorithms (Similar to Market Makers in the past)

National Stock Exchange of India is one of the leading exchanges in the world on several key parameters. Technology at the exchange helps to fulfill the demand for capacity, reliability

and performance ensuring the competitive edge of NSE.

Table 2: NSE's Global Rankings

Parameters	Rank
Single Stock Options	8th
Single Stock Futures	2nd
Index Options	1st
Index Futures	7th

Source: WFE (Rankings done for the period Jan-Dec 2014)

Core Trading System:

NSE's trading system, called National Exchange for Automated Trading (NEAT), is a state of-the-art client server based application. It has uptime record of over 99% with latency is in single digit millisecond level for all orders entered into the NEAT system.

Earlier generation of trading system was highly dependent on the growth of microprocessor industry for improved scalability. This was creating speed breakers in the growth demanded by Indian market participants. Some time back NSE re-architected the trading system to achieve unlimited scalability. The system now has a multi layer architecture is designed for unlimited scalability at every layer. Each layer of trading system can be scaled up by adding more hardware to the layer. The re-architecting of the system has eased out meeting the ever growing capacity needs of Trading. This application extensively uses in-memory database technology to provide for performance needs expected from a Matching system. The matching engine response time can be measured in single digit millisecond for the thousands of transactions processed by the system every second. To complement the matching engine speed, Market Data is generated and distributed at a very high refresh frequency. Using the Multicasting the market data access is accessible to all trading members almost simultaneously.

NSE SCHEMATIC

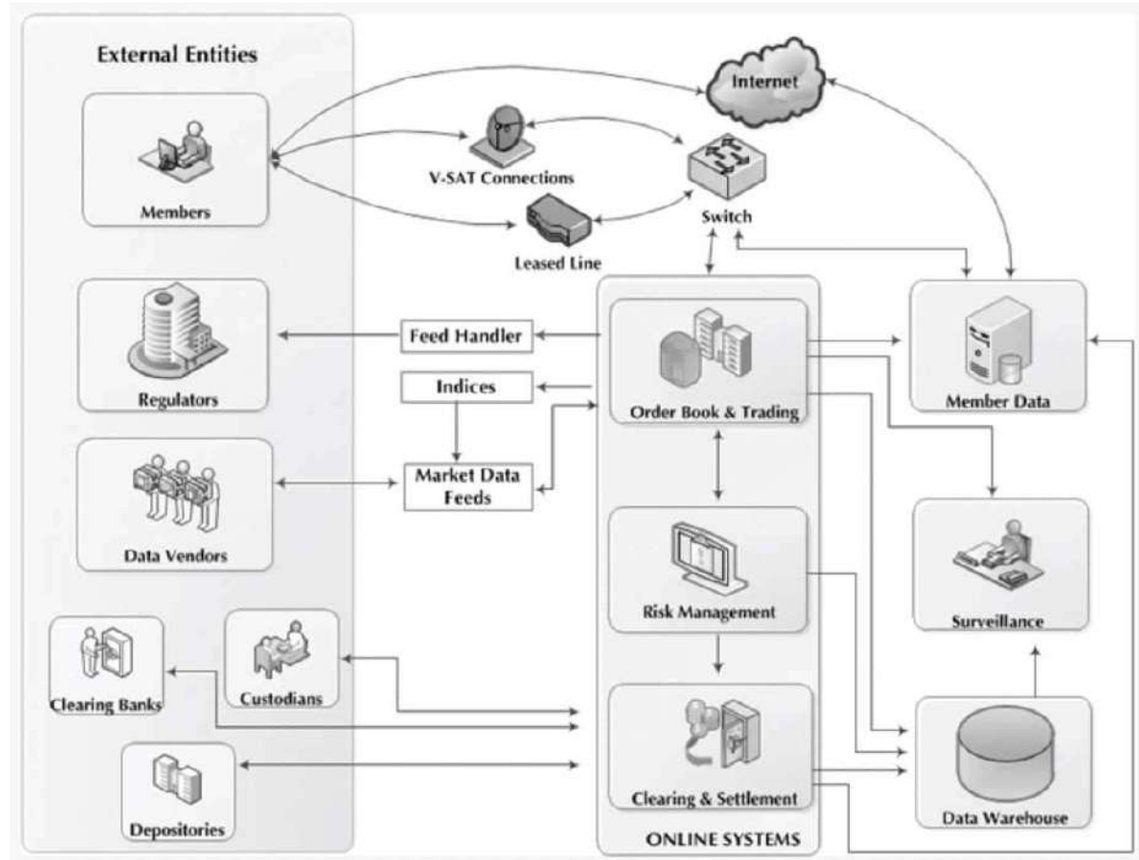

Figure 1: NSE Trading System Block Diagram

Source: NSE website

4.1) **ALGO TRADING:**

Algorithmic trading refers to use of computers and programs to generate and execute orders in markets with electronic access. Any order that is generated using automated execution logic shall be known as algorithmic trading. AT refers to orders that are automatically placed in the market by software programs built on certain mathematical models. AT involves the translation of public information into mathematical models that compute orders that are placed automatically on the market for execution. AT represents a fusion of human traders and computers where the role of the humans input shifts away from executing trades to instructing the computer on how to place buy/sell trades. Information today travels at a rate faster than anyother time in recorded human history,and therefore those who utilize this faster pace of transferof transformation could benefit in their trading operations. The usage of

computers for sending an order to the exchange based on predefined rules is broadly referred to as algorithmic trading. It is based on technology driven pre-programmed mathematical model-based stock trading.The growth of algorithmic trading is very closely related to the development of the electronic trading market. The important mile stones in the global electronic trading industry are tabulated below.

Table 3: Milestones in Global Electronic Trading

Year	Event
1969	Instinet allowed electronic block trading
1971	NASDAQ electronic board started
1972	Cantor establishes first electronic market place for US Government Securities
1976	NYSE DOT routes small orders
1978	ITS links NYSE and other US Stock Exchanges
1980	Instinet enables DMA to US exchanges
1982	Tokyo Stock Exchange introduces computer assisted order routing and execution system
1986	Paris Bourse introduced an electronic trading system
1988	MTS Platform creates secondary market for Italian government bonds
1992	CME launches first version of GLOBEX electronic futures platform
1993	EBS adds competition for spot FX
1997	SEC Changes order handling rules
1998	Eurex launches first fully electronic exchange for futures
2001	Liquidnet ATS created allowing "dark pool" buy side crossing for equities
2007	US regulation NMS, European MiFID (Markets in Financial Instruments Directive) regulation comes into force

Table 4: AT Milestones in INDIA

Year	Event
Apr,2008	SEBI allows Direct Market Access
Aug,2010	Smart Order Routing introduced
Mar,2012	SEBI issues broad guidelines for algorithmic trading
Jan.2013	Forward Market Commission (FMC) issues guidelines for algorithmic trading in commodities.

Direct Market Access (DMA)

Direct Market Access is a recent innovation in Indian Securities Markets. DMA and Algorithmic Trading are popularly used in the advanced markets of US, UK and Europe. DMA provides large investors & institutions with great control over trades, faster and better quality of execution, facilitation of complex trading strategies, lowered brokerage fees. So the transaction flows through DMA have shown significant increase. DMA facility was not available in the Indian Markets till 2008. SEBI permitted stock exchanges to facilitate its trading members to offer DMA to institutional clients in April 2008. Foreign Institutional Investors and Domestic Institutions such as mutual funds and insurance firms can now directly execute their trading without any manual intervention by their brokers.

CoLocation

The advent of automated trading systems (ATS) and High-Frequency Trading Systems (HFT) led to a need for low latency trading architectures. One of the ways to reduce latency in a very big way is to move physically closer to the exchange. That means the signal generated by the order manager for an order reaches the exchange faster. To cater to the requirement,co-location facilities (rack space for the trading server within the exchange) are offered. NSE introduced co-location services for the traders in Jan 2010. This facility allows market participants to rent servers located within the NSE's premises. Co-Location refers to exchanges allowing members to setup automated trading systems (ATS) on their premises to reduce latency i.e. time required for orders to flow between the exchange and the broker's trading system. NSE on an average saw 22% of its turnover from co-location servers during 2015.

INTERNET BASED TRADING (IBT)

Internet Based Trading is a method of trading in securities whereby it is possible for the investors to do trading through internet. It is also called online trading. Internet has enabled clients to transact online through the brokerage's website and execute the trade themselves. This significantly reduces the number of contact points required otherwise, thus reducing cost and load (of servicing clients) on stock broking companies branches. Online trading a popular notion among retail investors is one of the most successful industries created by the internet revolution. Internet Trading started in India on 1st April 2000 with 79 members seeking permission for IBT. On 1st Feb 2000, NSE opened up the IBT for its members. The SEBI committee on Internet based securities trading and services has allowed the net to be used as an order routing system (ORS) through registered stockbrokers on behalf of their clients for execution of transactions.

MOBILE TRADING

Mobile Trading is a new channel approved for trading enables brokers to provide trading capability on smart phones and mobile devices. Both internet trading and mobile trading has enabled the retail investor to track the stock market movements closely. The mobile trading is a flexible,dynamic and interactive facility which enables the investors to trade in the market even on the move.

SMART ORDER ROUTING (SOR)

SEBI has approved SOR facility since August 2010. SOR allows brokers trading engines to systematically choose the execution destination based on the factors like price, cost, speed, likelihood of execution and settlement, size, nature or any other consideration relevant to execution of the order. India is among the very few markets in Asia where the same stocks are listed on two different exchanges. When an investor wants to trade a security, he would want to execute at the best possible price, which at times would be on the NSE and at others on the BSE. With the advancement of electronic trading in the country, an increasing proportion of orders are being sent to exchanges by automated systems. SOR technology enables these systems to choose the best possible price, just like their human trading desk counterpart, but at machine speed.

5. MODE OF TRADING AT NSE

Statistics related to Internet Based trading at NSE is given below.

In the Cash Market Segment Internet Trading Value in increased from Rs 7, 65,271 Cr to 10, 05,984 Cr during 2010-11 to 2014-15. The IBT value as a % of trading volume increased from 10.70% to 11.62%.

Table 5: Internet Trading Value in the CM segment in comparison with total trading volumes at NSE.

Year	Enabled Members*	Registered Clients*	Internet Trading Value (` Crores)	Internet Trading Value (US $ bn)	% of total trading volume
2010-11	387	56,40,513	7,65,271	171.39	10.70
2011-12	428	61,48,447	5,97,430	116.78	10.63
2012-13	445	62,68,798	5,83,073	107.20	10.76
2013-14	459	68,74,574	6,27,478	105.00	10.99
2014-15	480	63,59,312	10,05,984	160.72	11.62

Source: NSE Fact Book (2015)

Table 6: Internet Trading in the F&O Segment of the Exchange

Year	Enabled members *	Registered clients *	Internet Trading Value (` Crores)	Internet Trading Value (US $ bn)	% of Total Trading Volume
2010-11	380	43,63,400	43,18,184	967.12	7.38
2011-12	421	41,51,677	49,10,582	959.91	7.83
2012-13	437	38,33,149	59,05,158	704.76	9.36
2013-14	451	41,16,039	76,52,572	688.78	10.01
2014-15	472	46,95,396	1,16,13,948	1,855.54	10.44

Source: NSE Fact Book (2015)

In the F&O Market Segment Internet Trading Value in increased from Rs 43,18,184 Cr to 1,16,13,948 Cr during 2010-11 to 2014-15. The IBT value as a % of trading volume increased from 7.38% to 10.44%.

Table 7: NSE Cash Market – Mode of Trading(Jan – Dec 2015)

Mode Of Trading	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Non-Algo	45	44	44	44	43.9	41.28	44.79	40.09	39.28	42.74	41.90	45.71
Algo	16	16	18	18	3.95	17.74	16.06	18.55	18.28	16.63	19.48	16.11
Direct Market Access(DMA)	1	0	1	1	4.02	0.93	0.47	0.49	0.63	0.53	0.66	0.38
Co-location(CoLo)	23	25	23	23	35.98	24.60	22.45	25.95	25.66	22.9	22.02	20.73
Internet Based Trading(IBT)	11	11	11	11	11.17	11.03	11.71	10.47	11.3	12.11	11.18	12.32
Mobile Trading	1	1	1	1	0.81	1.63	1.76	1.75	2.16	2.39	2.27	2.62
Smart Order Routing(SOR)	3	3	2	2	0.18	2.79	2.76	2.70	2.69	2.71	2.50	2.12

Source: NSE Website

Table 8:NSE Derivatives Market (JAN –DEC 2015)

Mode Of Trading	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Non-Algo	45	44	44	44	43.9	41.28	44.79	40.09	39.28	42.74	41.90	45.71
Algo	16	16	18	18	3.95	17.74	16.06	18.55	18.28	16.63	19.48	16.11
Direct Market Access(DMA)	1	0	1	1	4.02	0.93	0.47	0.49	0.63	0.53	0.66	0.38
Co-location(CoLo)	23	25	23	23	35.98	24.60	22.45	25.95	25.66	22.9	22.02	20.73
Internet Based Trading(IBT)	11	11	11	11	11.17	11.03	11.71	10.47	11.3	12.11	11.18	12.32
Mobile Trading	1	1	1	1	0.81	1.63	1.76	1.75	2.16	2.39	2.27	2.62
Smart Order Routing(SOR)	3	3	2	2	0.18	2.79	2.76	2.70	2.69	2.71	2.50	2.12

Source: NSE Website

Table 9:Mode of Trading in the cash segment at NSE

Month	Algo	Non-Algo	DMA	Co-Location	IBT	Mobile	SOR
Apr-2010	10.7	76.4	0.3	2.3	10.4	0.0	0.0
Apr-2011	16.3	64.8	0.7	7.9	10.2	0.0	0.1
Apr-2012	16.2	56.6	0.5	14.2	11.2	0.2	1.0
Apr-2013	18.8	50.1	1.0	16.1	10.9	0.5	2.6
Apr-2014	17.0	49.0	0.8	18.7	11.6	0.9	2.0
Dec-2014	15.8	46.6	0.3	22.6	11.3	1.2	2.2

Source: NSE Fact Book (2015)

NSE Cash Market

Algo Trading increased from 10.7 % in Apr 2010 to 15.8% in Dec 2014. It recorded a high of 20.9% in Aug 2013. Non-Algo trading decreased from 76.4% in Apr 2010 to 46.6% in Dec2014. DMA share has stagnant at 1.3% during 2010-14. Co-Location share has increased from 2.3% in Apr 2010 to 22.6% in Dec 2014. IBT is almost steady at 10.4% in Apr 2010 –Dec 2014. Mobile Trading has remained marginal at around 1.2% in Dec 2014. SOR is also insignificant at 2.2% in Dec 2014.

Table 10:Mode of Trading in the Equity Derivatives Segment at NSE

Month	Algo	Non-Algo	DMA	Co-Location	IBT	Mobile	SOR
Apr-2010	7.6	80.5	2.0	1.8	8.0	0.0	0.0
Apr-2011	19.9	67.1	2.7	2.2	8.0	0.0	0.1
Apr-2012	4.3	59.0	7.6	18.8	10.3	0.1	0.0
Apr-2013	5.4	50.8	8.9	24.7	10.0	0.2	0.0
Apr-2014	4.4	46.1	8.1	29.8	11.2	0.5	0.0
Dec-2014	3.4	46.7	3.9	34.0	11.3	0.6	0.0

Source: NSE Fact Book (2015)

Algo Trading increased from 7.6 % in Apr 2010 to 26.3% in Mar 2012 but it decreased to 3.4% since then up to Dec 2014. Non-Algo trading decreased from 80.5% in Apr 2010 to 46.7% in Dec2014. DMA share has stagnant at 4 % during 2010-14. Co-Location share has increased from 1.8% in Apr 2010 to 34% in Dec 2014. IBT increased from 8% in Apr 2010 to 11.3% in Dec 2014. Mobile Trading has remained marginal at around 0.6%. SOR is also insignificant during Apr 2010- Dec 2014.

7. FINDINGS & CONCLUSIONS

6.1) Algorithmic Trading has rapidly become a significant portion of Indian Markets. Over 200 NSE Members utilize algos and constitute around 37-38% of overall values in 2015.

6.2) Several algos are operated from co-location facilities at Exchanges. At NSE co-location, more than 100 brokers are operational and constitute around 22% of total volume.

6.3) In the Cash Market Segment Internet Trading Value in increased from Rs 7,65,271 Cr to 10,05,984 Cr during 2010-11 to 2014-15. The IBT value as a % of trading volume increased from 10.70% to 11.62%.

6.4) In the F&O Market Segment Internet Trading Value in increased from Rs 43,18,184 Cr to 1,16,13,948 Cr during 2010-11 to 2014-15. The IBT value as a % of trading volume increased from 7.38% to 10.44%.

6.5) NSE Cash Market

Algo Trading increased from 10.7 % in Apr 2010 to 15.8% in Dec 2014. It recorded a high of 20.9% in Aug 2013. Non-Algo trading decreased from 76.4% in Apr 2010 to 46.6% in Dec2014. DMA share has stagnant at 1.3% during 2010-14. Co-Location share has increased from 2.3% in Apr 2010 to 22.6% in Dec 2014. IBT is almost steady at 10.4% in Apr 2010 –Dec 2014. Mobile Trading has remained marginal at around 1.2% in Dec 2014. SOR is also insignificant at 2.2% in Dec 2014.

6.6) NSE Equity Derivatives Market

Algo Trading increased from 7.6 % in Apr 2010 to 26.3% in Mar 2012 but it decreased to 3.4% since then up to Dec 2014. Non-Algo trading decreased from 80.5% in Apr 2010 to 46.7% in Dec2014. DMA share has stagnant at 4% during 2010-14. Co-Location share has increased from 1.8% in Apr 2010 to 34% in Dec 2014. IBT increased from 8% in Apr 2010 to 11.3% in Dec 2014. Mobile Trading has remained marginal at around 0.6%. SOR is also insignificant during Apr 2010- Dec 2014.

7. LIMITATIONS OF THE STUDY

This study considers secondary data available from Indian stock exchange websites and does not consider the high frequency data

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